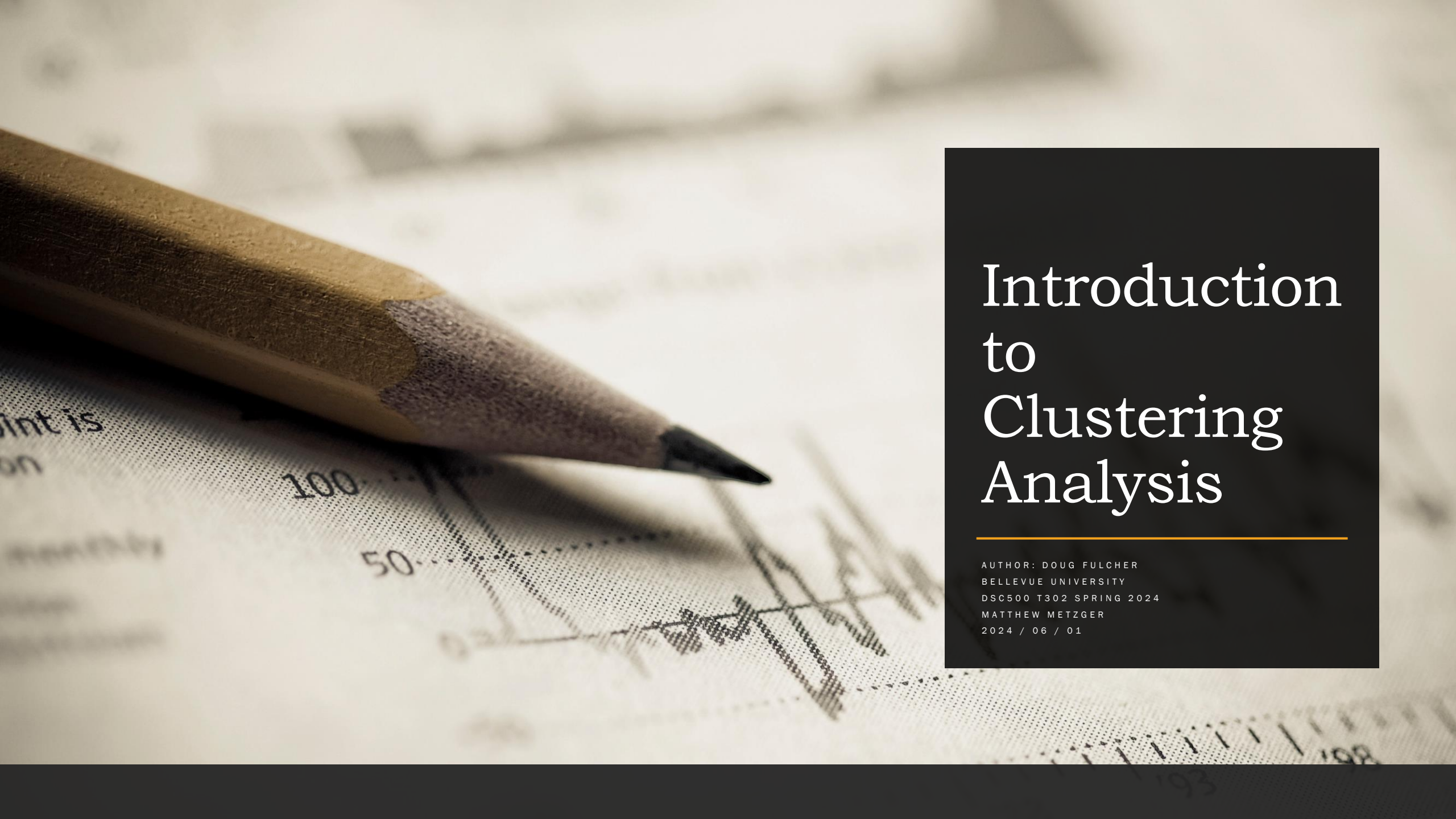




# Introduction to Clustering Analysis

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2024 / 06 / 01



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# AGENDA

INTRODUCTION

CLUSTERING

APPLICATIONS

CONCEPTS

TOOLS

CONCLUSION



# What is Clustering?

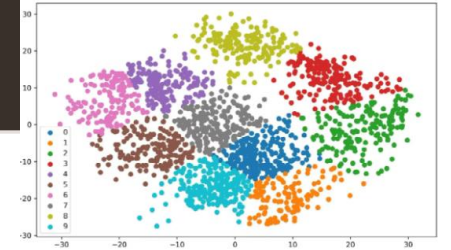
Clustering allows us to group objects such that items similar to each other are in the same group (called a cluster).

Clustering allows us to discover structure, patterns, and groupings within data that can be further analyzed.

Example: clustering whiskeys based on similar characteristics (Color, Nose, Body, Palate, and Finish) (Provost & Fawcett, 2013).

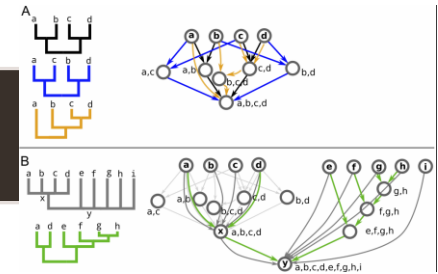
## K-Means

- K represents the number of clusters created
- Each object belongs to the cluster with the nearest mean



## Hierarchical

- Tree of Clusters, Bottom-up or top-down
- Nearby objects are more related than objects far away

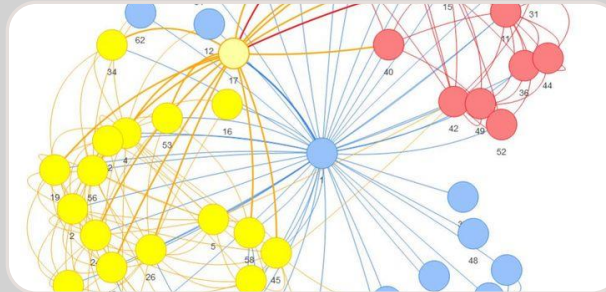


# Real World Applications



## Risk-based Segmentation

- Insurance companies comparing customers to determine risk and assess pricing



## Social Networking

- Creates clusters based on the degrees of separation from one user to another based on their “friends”

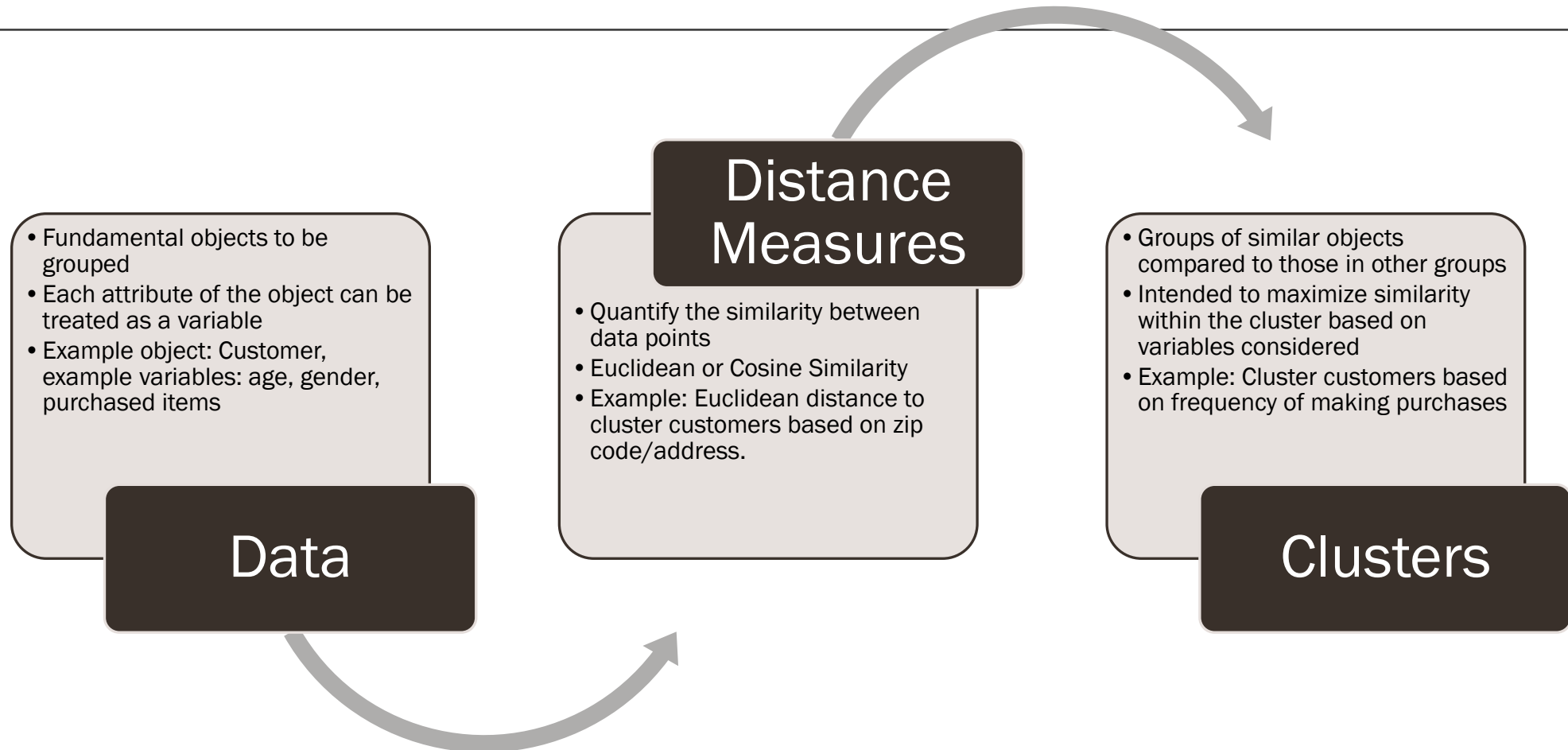


## Healthcare Results

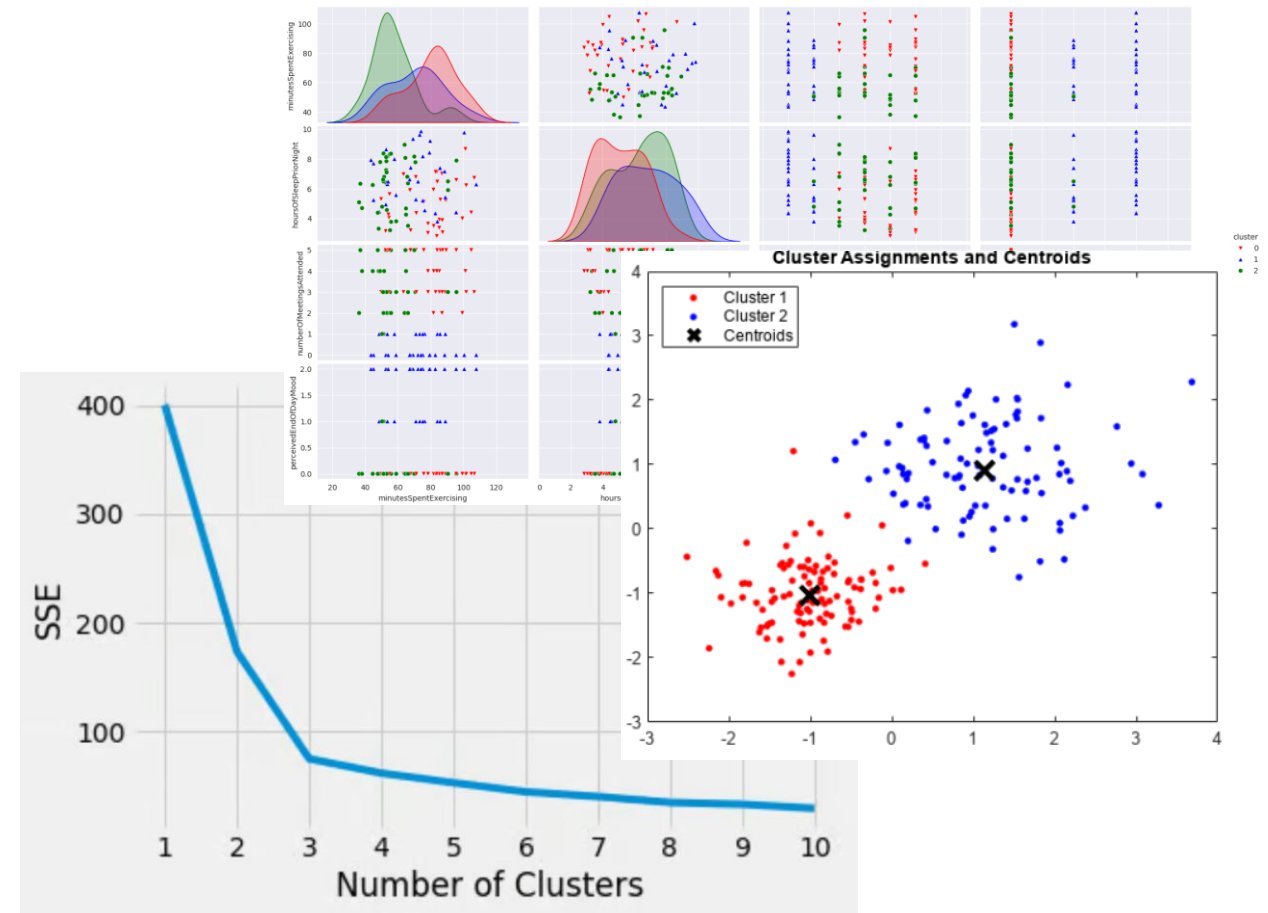
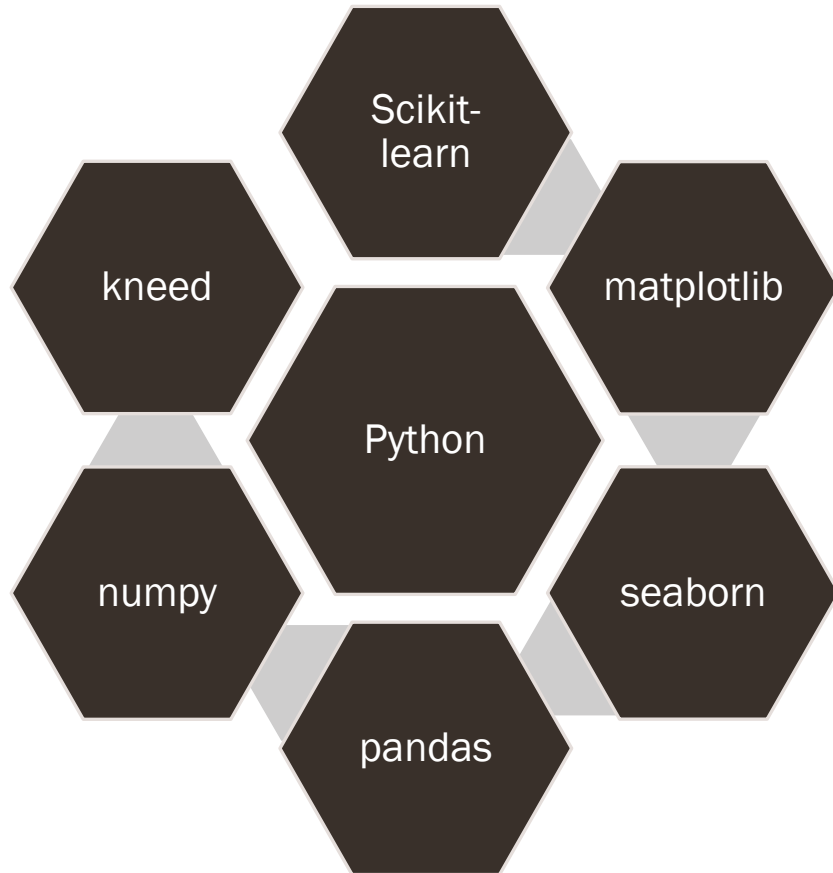
- Using image recognition software, medical scans (e.g., xrays, MRIs, etc) can be analyzed to cluster patients whose images present a likelihood of diagnoses

# Key Clustering Concepts

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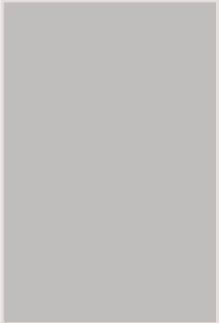


# Clustering Tools & Libraries

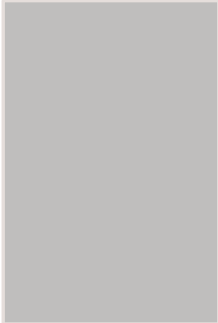


# Challenges

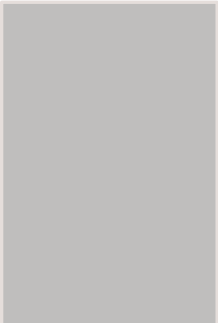
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Number of  
Clusters



Interpretability  
of Clusters



Scaling over  
datasets

# Summary / Q&A

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Clustering allows us to analyze groups of populations that are similar in nature.

Through clustering, we can quickly identify patterns in our data that may not otherwise be apparent.

Many real-world applications for clustering can drive positive business and societal outcomes.

Clusters are formed by evaluating the distance between variables for records in a data set. The shorter the distance, the more likely the similarity.

There are many tools and libraries that can help with determining clusters and visualizing clusters.

As our datasets grow, we must evaluate our clustering models and available resources (compute, storage) or adjust the models to target specific populations of interest.